

Python Project Setup with UV & VS Code

Windows & macOS — Step-by-Step Setup Guide

This guide walks you through creating a Python project in VS Code, installing the uv package manager, creating a virtual environment, installing libraries from requirements.txt, and verifying the setup with a Jupyter Notebook.

1. Create a Project Folder

Create a new folder anywhere convenient on your computer. For example:

AI_Project

This folder will be the main project directory containing your Python environment, project files, requirements.txt, and notebooks.

2. Open the Folder in VS Code

1. Open Visual Studio Code.
2. Go to File → Open Folder.
3. Select the project folder you created.
4. Click Select Folder / Open.

3. Install uv Package Manager

If uv is already installed, you can skip the installation step. Otherwise, use the official uv installation method for your operating system.

Windows: *powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"*

MacOS: *curl -LsSf https://astral.sh/uv/install.sh | sh*

4. Completely Close VS Code

After installing uv, completely close VS Code. This helps VS Code and its integrated terminal pick up any PATH changes made during the uv installation.

5. Reopen VS Code and Verify uv

5. Reopen VS Code.
6. Open the project folder again.
7. Open the integrated terminal using Terminal → New Terminal.
8. Run the following command:

uv

If uv is installed correctly, the terminal should display uv's command-line help/version information.

6. Initialize the uv Project

Make sure the terminal is currently opened inside your project folder, then run:

```
uv init
```

This initializes the project and creates project configuration files such as pyproject.toml.

7. Create the Virtual Environment

Create a project-specific virtual environment with:

```
uv venv
```

By default, uv creates the environment in a .venv folder inside the project directory.

8. Activate the Virtual Environment

Windows — PowerShell:

```
.venv\Scripts\Activate.ps1
```

Windows — Command Prompt (cmd):

```
.venv\Scripts\activate.bat
```

macOS / Linux:

```
source .venv/bin/activate
```

After activation, you should normally see (.venv) at the beginning of the terminal prompt.

9. Create requirements.txt

In the VS Code Explorer, create a new file named exactly:

```
requirements.txt
```

Add the required libraries, one package per line:

```
transformers  
numpy  
pandas  
torch  
torchvision
```

torchaudio
ipykernel

10. Install the Libraries

With the virtual environment activated, install all packages listed in requirements.txt:

```
uv pip install -r requirements.txt
```

Wait for the installation to finish. PyTorch packages can take some time and may require additional disk space.

11. Create a Jupyter Notebook

9. In the VS Code Explorer, create a new file.
10. Name it test.ipynb.
11. If prompted to install the Jupyter extension, install it.
12. Select the Python interpreter associated with your .venv environment as the notebook kernel.

12. Test the Setup

In the first notebook cell, enter:

```
print("anant")
```

Run the cell. The expected output is:

```
anant
```

13. Final Project Structure

```
AI_Project/  
├── .venv/  
├── .python-version  
├── pyproject.toml  
├── requirements.txt  
└── test.ipynb
```

Note: uv may create additional project files depending on the uv version and project configuration.

Quick Command Summary

```
uv  
uv init  
uv venv
```

```
# Activate — Windows PowerShell  
.venv\Scripts\Activate.ps1
```

```
# Activate — macOS  
source .venv/bin/activate
```

```
uv pip install -r requirements.txt
```

Official References

uv Installation: <https://docs.astral.sh/uv/getting-started/installation/>

uv Project Initialization: <https://docs.astral.sh/uv/concepts/projects/init/>

uv Project Guide: <https://docs.astral.sh/uv/guides/projects/>

Setup complete — your Python environment is ready for development.